

Assessment No. 3
Error Detection and Correction Code

Registration No. _____

Dated _____

Practical significance:

Error detection and correction code plays an important role in the transmission of data from one source to another. The noise also gets added into the data when it transmits from one system to another, which causes errors in the received binary data at other systems. The bits of the data may change (either 0 to 1 or 1 to 0) during transmission.

It is impossible to avoid the interference of noise, but it is possible to get back the original data. For this purpose, we first need to detect either an error is present or not using error detection codes. If the error is present in the code, then we will correct it with the help of error correction codes. The error detection codes are the code used for detecting the error in the received data bitstream. In these codes, some bits are included appended to the original bitstream.

Error detecting codes encode the message before sending it over the noisy channels. The encoding scheme is performed in such a way that the decoder at the receiving can find the errors easily in the receiving data with a higher chance of success.

- These codes are used when we use message backward error correction techniques for reliable data transmission. A feedback message is sent by the receiver to inform the sender whether the message is received without any error or not at the receiver side. If the message contains errors, the sender retransmits the message.

- In error detection codes, in fixed-size blocks of bits, the message is contained. In this, the redundant bits are added for correcting and detecting errors.

- These codes involve checking of the error. No matter how many error bits are there and the type of error.

- Parity check, Checksum, and CRC are the error detection technique.

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AIM

Implement the technical concepts of various error detecting and correcting coding schemes using Python.

Source Code

Execution Scenario [Sample Output]

Review Questions

1. Identify few significance of Even and Odd Parity.
2. Write the use of CRC.
3. Name the standard of CRC.
4. Write the working scenario of generator polynomial.